



ALVIN A S
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CGPA : 9.14



ACADEMIC DETAILS

COURSE	INSTITUTE/COLLEGE	BOARD/ UNIVERSITY	SCORE	YEAR
CLASS XII	Christu Jyothi English Medium School, Chullimanoor, Nedumangadu, Thiruvananthapuram	CBSE	97.2 %	2023
CLASS X	Christu Jyothi English Medium School, Chullimanoor, Nedumangadu, Thiruvananthapuram	CBSE	97 %	2021

Subjects / Electives	Professional Ethics, Programming Laboratory, Database Management System, Professional Communication, Physics of Materials, Logic Design, Data Structure and Algorithms, Biology For Engineers, Data Structure and Algorithms Laboratory, Computer Organization, Mathematics, Innovation & Entrepreneurship, Discrete Structures, Computer Programming, Hardware Laboratory, Computer Graphics and Product Modelling, Operating System, Systems Programming Lab, Introduction to Computing Science, Program Design, Program Design Laboratory, Theory of Computation
Technical Proficiency	Application Development, Autocad 3D, Java, Web Development, C Programming, Dart, Node.js, Verilog, DSA, React.js, PostgreSQL, Blender, Figma, JavaFX, Python, Flutter, Express.js

SUMMER INTERNSHIP / WORK EXPERIENCE

Question Paper Preparation, Pediaverse	May 2024 - Aug 2024
Created chapter-wise question papers for JEE, NEET, and KEAM aspirants in Physics, Chemistry, and Mathematics. Curated questions of varying difficulty levels to align with competitive exam standards. Received a stipend of 3000 for the internship.	

PROJECTS

DCode - App Development (Java, JavaFX)	Jan 2025 - Apr 2025
Developed a specialized Java code editor named DCode, aimed at enhancing the coding experience for learners and developers through a clean interface and a rich set of integrated tools. Built using Java and JavaFX, the application combines conventional editor features with unique visualizations and educational utilities.	
DCode offers an intelligent syntax-aware editor with features like auto-indentation, bracket completion, and real-time syntax highlighting tailored for Java. It includes a built-in terminal for compiling and running Java code seamlessly within the application.	
A standout feature of DCode is the Objects Pane, which visually represents static variables (including arrays and 2D arrays) as interactive nodes. These structures are rendered with contextual analysis such as cycle detection and bipartiteness, enhancing conceptual understanding of data structures. The Message Box further supplements this by displaying metadata and analytical insights about selected objects.	
The application supports multi-file editing, a customizable abbreviation system (e.g., sout !' System.out.println()), a file explorer, and a pop-up documentation window that provides in-app learning resources on data structures and programming languages.	
Key Features:	
- Java-aware code editor with auto-complete (work in progress), syntax and error highlighting. - Static object visualization with structural analysis (e.g., graph cycles, bipartiteness, min and max heaps). - Integrated terminal for compiling and running Java programs. - Customizable themes and editor behavior. - In-app documentation on programming topics and data structures. - Support for multiple open files with tab-style navigation. - Roadmap includes support for non-static variable visualization and plugin integration.	
Technologies Used: Java, JavaFX	
HashCure - Web Development (React, Express, PostgreSQL, Node)	Jun 2025 - Present
HashCure is a full-stack hospital management web application built using React, Node.js, Express, PostgreSQL, and Socket.IO, designed to streamline hospital operations, patient-doctor interaction, and healthcare data management. The platform supports four user roles: Patient, Doctor, Hospital Admin, and Super Admin, offering tailored features and workflows for each.	

Key Features by User Role

Patients:

- Hospitals Page: Search and filter hospitals by district, specialty, and ownership (public/private).
- Doctors Page: Filter doctors by location, hospital, specialization, availability, gender, and fees.
- Appointments Page: View and filter appointments by status (scheduled, cancelled, pending) and date (today, tomorrow, upcoming, past).
- Donors Page: Find willing blood donors filtered by blood group, age, and BMI (excludes recent donors).
- Hospital Page: View hospital details, rate and comment, and explore in-hospital doctors.
- Doctor Page: Explore doctor profiles with details, ratings, experience, YouTube videos, and comment functionality.
- Take Appointment Page: Schedule an appointment via a slot picker with real-time availability and purpose specification.
- Reports Page: Add disease-specific logs with graphical history, upload medical reports, and manage medication details.
- User Profile: Update basic info (DOB, address, height, etc.), manage allergies, view and filter appointment history.
- Notifications Page: Receive real-time notifications for appointment updates or admin alerts.
- Settings Page: Toggle blood donor status, donation frequency, last donation date, and theme preferences.
- View Appointment Details Page: In-depth appointment summary including diagnosis, treatment plan, and complaints.

Doctors:

- Doctor Appointments Page: Filter and manage appointments, accept/reject slots, and initiate consultations.
- Patients Page: View and filter assigned patients by name or age.
- Consult Patient Page: View history and fill in current appointment details (chief complaint, diagnosis, treatment). Book follow-ups directly.
- Doctor Profile Page: Set bio, working hours, hospitals, manage video links, and respond to patient comments.
- Reports & Appointments Access: Full access to patients' records and appointment histories.
- Doctor Settings Page: Same as patients with added "Allow Direct Appointment" toggle.

Hospital Admin:

- Doctors Management: View, add, or remove doctors from the hospital.
- Patients Management: View all patients associated with the hospital.
- Appointments Viewer: Select and view appointment records of any patient.
- Reports Viewer: Access detailed medical reports of patients.
- Hospital Profile Management: Update hospital image, specialization, working hours, ownership, and view public comments.

Super Admin (Me):

- Full control over the platform's data and user roles (add hospitals, view data)

Technical Overview

- Frontend: React
- Backend: Node.js + Express
- Database: PostgreSQL
- Real-time: Socket.IO (for notifications and appointment updates)

Project Highlights

- Real-time appointment booking and notifications
- Role-based UI with customized feature sets
- Integrated blood donor finder and filtering
- Graph-based health tracking with disease-specific visualizations
- Rich doctor and hospital profiles with rating/commenting system
- YouTube integration for doctor-shared video content
- All data accessible to doctors for better consultation context
- Mobile-friendly and user-centered interface design

Project Status:

- Backend: Completed
- Frontend: Under development
- Project Type: Individual

Jobly - Web Development (React, Express, PostgreSQL, Node) Jun 2025 - Present

Jobly is a professional social networking platform modeled after LinkedIn, built using React, Node.js, Express, PostgreSQL, and Socket.IO. Designed for both employers and employees, the platform enables users to build professional connections, share career-related content, and engage in real-time recruitment workflows.

Users can register as either employers or employees, with distinct capabilities. Employees can create posts showcasing their skills, certifications, blogs, or project screenshots, which others can like, comment on, or share. Users can follow and connect with others, and a

personalized feed is generated based on declared interests and skills set during onboarding, which can be updated later in the profile settings.

The platform supports dual profile views: one for viewing one's own profile with edit capabilities, and another for viewing other users' public profiles showcasing their skills, posts, and connections. A comprehensive search and discovery system allows users to find others by name, domain expertise, or post categories.

The integrated real-time messaging system, powered by Socket.IO, enables direct chat between users with support for sending, editing, deleting, and pinning messages. Notifications are pushed for actions such as likes, comments, connection requests, and job responses.

Employers can post job vacancies by specifying required skills, experience, and job details (description, salary, etc.). Matching employees receive real-time invitations, and can respond with "Interested" or "Not Interested". Employers can then review interested candidates, accept or reject applications, and manage shortlisted applicants in a dedicated selection panel, which includes email contact options.

- Additional features include:
- Create Post page for content sharing
- Friends Page with tabs for followers, following, and mutual connections
- Settings Page for privacy controls and theme selection (light/dark mode)
- Scalable notification system for all major interactions

Key Features:

- Role-based account system (Employer/Employee)
- Skill- and interest-driven content feed
- Real-time messaging and job invitation via Socket.IO
- Post creation and engagement (like, comment, share)
- Connection system with suggestion engine
- Employer dashboard for posting jobs and managing candidates
- Notification and profile management modules
- Privacy settings and theming options

Technologies Used: React, Node.js, Express, PostgreSQL, Socket.IO

Team Project Status:

Backend development completed; frontend implementation in progress as a collaborative team project.

HashChats - Web Development (React, Express, PostgreSQL, Node)

May 2025 - Jun 2025

Developed HashChats, a full-featured, real-time communication platform built using React, Node.js, Express, PostgreSQL, Socket.IO, and WebRTC. Designed to replicate the experience of modern messaging platforms, HashChats supports private chats, group conversations, media sharing, and real-time audio/video calls.

The application provides a secure authentication system with session persistence, allowing users to remain logged in across visits. The responsive user interface features a navigation sidebar for quick access to chats, groups, favorites, profile settings, and logout functionality.

Users can send, edit, delete, reply to, pin, and copy messages, with support for sharing images, videos, and documents. All interactions—including new messages and message updates—are synchronized in real-time using WebSocket communication via Socket.IO.

The platform includes robust 1-on-1 and group chat capabilities, with presence indicators (last seen/active), group creation tools, and media history views. Built-in audio/video calling is powered by WebRTC, offering seamless peer-to-peer communication with minimal setup.

A dedicated chat details panel provides contextual information, including user details, group members, and a scrollable media archive. The user profile management system enables avatar changes, username and description editing, and quick media access.

Key Features:

- Secure user authentication with session persistence
- Real-time 1-on-1 and group messaging
- Advanced message actions: edit, delete, reply, pin, copy
- Multimedia sharing: images, videos, PDFs
- Integrated audio and video calling via WebRTC
- Real-time data synchronization using Socket.IO
- Profile customization and media history browsing
- Scalable PostgreSQL backend with relational data design

Technologies Used:

- Frontend: React
- Backend: Node.js, Express
- Database: PostgreSQL

- Real-time Communication: Socket.IO
- Audio/Video Calls: WebRTC

Hash-Movies - Web Development (React, Express, MongoDB, Node)

May 2025 - May 2025

Developed a full-stack movie discovery platform called HashMovies, built using React for the frontend, Node.js and Express for the backend, and MongoDB for data storage. The platform is designed to help users search, explore, and engage with movies across a wide range of genres, languages, actors, and streaming platforms.

The application provides a robust search and filter system, enabling users to combine filters using both AND and OR logic—for instance, viewing movies that are both Action and Thriller, or either Action or Thriller. Users can personalize their experience by selecting their favorite genres and languages, which will be used to populate their home page recommendations. Additional curated sections such as Trending and Featured Movies enhance discoverability.

Each movie page displays comprehensive information including ratings, cast, genre tags, synopsis, and similar movies. Users can also comment on movies, providing social insight into film quality and content. Actor-specific pages offer biographical details, debut works, and notable performances, while dedicated genre and language pages showcase categorized movie collections and descriptions.

An integrated Admin Dashboard provides tools for managing platform content. Admins can view and edit sortable tables of users, movies, genres, and platforms, and perform CRUD operations (Create, Read, Update, Delete) on movies, genres, and platforms. Cloudinary is used for efficient and scalable image hosting. The website is hosted on Render.

Key Features:

- Advanced movie search with multi-parameter filtering using AND/OR logic
- Personalized home page based on user-preferred genres and languages
- Rich movie detail pages with cast, ratings, storyline, and recommendations
- Actor, genre, and language-specific pages for focused exploration
- User comments on movie pages for community feedback
- Admin dashboard with editable tables and content management capabilities
- Cloudinary integration for media asset storage

Technologies Used: React, Node.js, Express, MongoDB, Cloudinary

File Manager - App Development (Java, JavaFX)

Mar 2025 - Apr 2025

Designed and developed a modern desktop file manager using Java and JavaFX, emulating the behavior and usability of standard Linux file explorers. The application offers a clean, responsive UI and comprehensive file handling capabilities, delivering a smooth and intuitive user experience.

Users can efficiently navigate through directories and perform all essential file operations including create, rename, delete, move, and copy, using either mouse interactions or keyboard shortcuts. The file manager supports various view modes, including small, medium, large icons, and a tile view that displays detailed file metadata for better clarity.

A key usability feature includes drag-and-drop support, allowing users to easily move files and folders between directories. The UI is built to be adaptive and minimalistic, automatically adjusting layout and icon scaling based on the selected view mode.

Key Features:

- Full suite of file operations (create, rename, delete, move, copy)
- Multi-mode view options: small/medium/large icons and detailed tile view
- Drag-and-drop file handling between directories
- Responsive, user-friendly JavaFX interface
- Supports both mouse and keyboard-based interactions
- Have a Properties window like that in Windows Explorer

Technologies Used: Java, JavaFX, Java NIO/IO

Bus Ticket Management App - Web Development (React, Flask, MySQL, Node)

Feb 2025 - Apr 2025

Collaboratively developed a full-stack web-based bus ticket booking system using React for the frontend, Flask for the backend, MySQL for data storage, and Node.js for auxiliary services. This project simulates the functionalities of modern bus booking platforms such as RedBus, providing an intuitive user experience for both passengers and administrators.

The application allows users to register or log in, select their source and destination locations, choose the date of travel, and search for available buses. Users can refine their search using multiple filters such as:

- Bus type: AC, Non-AC, Sleeper, Local
- Service type: Fast, Super Fast, Limited Stop
- Departure & Arrival Time Windows: Before 6 AM, 6–12, 12–6, After 6 PM

The app includes a seat selection interface that mimics the physical layout of a bus. Once a seat is booked, it is marked and becomes unavailable to other users, ensuring real-time reservation integrity.

The system features a comprehensive Admin Dashboard with the following modules:

- Buses Management: View, edit, and add new buses along with their routes and schedules.
- User Management: View registered users and their booking histories.
- Revenue Analytics: Visualize data through graphs, including metrics such as user growth, income trends, top-performing buses, and most active users.

Key Features:

- Full-stack architecture with robust separation of concerns.
- Real-time seat booking and availability tracking.
- Filter-rich search interface for ease of use.
- Admin portal with CRUD operations and revenue analytics.
- Responsive UI and scalable database design.

Technologies Used: React, Flask, Node.js, MySQL

Medical Reminder App - App Development (Flutter)

Sep 2024 - Dec 2024

Designed and developed a mobile application using Flutter, Dart, and Hive to help users effectively manage and track their medication schedules. The app allows users to input detailed information about their medications, including:

- Medicine name
- Dosage timing (e.g., before breakfast, after lunch, etc.)
- Medicine type (tablet, syrup, etc.)
- Duration of consumption (lifetime, specific number of days/months/years)
- Frequency (daily, weekly, biweekly, monthly)

To improve usability, the medications are visually categorized in the UI under daily, weekly, and monthly sections. Users can also manage a minimal profile page, which displays basic personal information such as name, age, and ongoing medication courses, along with the ability to update their profile.

A dedicated settings page allows users to configure custom notification times for each type of medicine timing (e.g., before breakfast), enabling timely and consistent reminders.

Key Features:

- Secure and efficient local data storage using Hive.
- Categorized medication organization for clear visual grouping.
- Customizable scheduling options for duration and frequency.
- Notification system aligned with user-defined timings.
- Simple and intuitive profile and settings management.

Technologies Used: Flutter, Hive, Dart

Password Manager - App Development (Flutter)

Aug 2024 - Sep 2024

Developed a secure password manager mobile application using Flutter for the frontend and Hive as the local NoSQL database. This application is designed for personal use and enables users to store and manage their credentials—such as usernames, passwords, and related information—for various websites and applications.

The app ensures data privacy by utilizing local storage with encryption, ensuring that all sensitive information remains securely stored on the user's device. Credentials are organized into customizable categories such as Work, Academics, or Personal, enhancing usability and organization.

• Key Features:

- Encrypted local storage using Hive for offline access and security.
- Categorization of credentials into user-defined sessions.
- Simple, intuitive UI for adding, updating, and retrieving credentials.
- Organized display of data for platforms like Google, ChatGPT, etc., with labels for email, password etc.

Technologies Used: Flutter, Hive, Dart

POSITION OF RESPONSIBILITY	
Junior Executive - GDSC App Development Team	Feb 2025 - Present
Executive - FOSS Cell Executive Member FOSS Cell, NIT Calicut	Jan 2025 - Present
Actively contributed to organizing FOSS Meet as part of the Hospitality Committee. Managed accommodation arrangements and coordination	

for attendees. Also assisted in setting up workshop environments and configuring computer systems for hands-on sessions, ensuring a smooth technical experience for participants.

EXTRA CURRICULAR ACTIVITIES

Kho Kho
I have been actively involved in Kho Kho since 2014, starting from my school days where I not only participated regularly but also served as team captain for two years. My leadership role helped me develop teamwork, strategic thinking, and decision-making skills. During college, I was selected for the Kho Kho team, reflecting my continued passion and commitment to the sport. I later chose to step back from competitive play to focus more on my academics, but my enthusiasm for the game remains strong.

AWARDS AND RECOGNITIONS

CERTIFICATIONS

CERTIFICATION	CERTIFYING AUTHORITY
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COMPETITIONS

CONFERENCES AND WORKSHOPS

TEST SCORES

TEST NAME	DATE OF EXAM	SCORE
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PATENTS

PUBLICATIONS

SCHOLARSHIPS

VOLUNTEER EXPERIENCE

NSS NITC - Role: Volunteer Cause: Social Services Volunteer National Service Scheme (NSS), NIT Calicut	Jan 2024 - Present
Actively participated in two annual NSS camps and various community outreach programs. Contributed to activities such as school infrastructure reinforcement, cleanliness drives, socio-economic surveys, and painting of pre-KG schools. Demonstrated commitment to social responsibility and rural development through hands-on volunteering efforts.	
Yuvasangham - Role: Volunteer Cause: Arts and Culture Volunteer Yuvasangham Cultural Exchange Program, NIT Calicut	Sep 2023 - May 2024
One of three core volunteers coordinating the entire program. Facilitated the visit of students from Odisha, managed accommodation, and organized cultural exchange activities. Designed brochures, posters, and flex banners, and ensured smooth execution of the event.	

LANGUAGES KNOWN

English, Malayalam, Hindi, Tamil